Week 04 Homework: Recurrence

1 Exercises

Exercise 1. A person needs to climb a staircase with n steps. At each step, the person can climb either 1 step or 2 steps. How many different ways are there for the person to climb to the top?

Exercise 2. How many ways are there to tile a rectangular floor of size $2 \times n$ using tiles of size 1×2 and 2×2?

Exercise 3. Solve the following recurrence relations:

a) $a_n - 2a_{n-1} = 6n$ for $n \geq 1$ with initial condition $a_0 = 2$.

b) $a_0 = 1, a_1 = 4$ and $a_n = 6a_{n-1} - 9a_{n-2}$ for $n \geq 2$.

Exercise 4. Find the general formula for the sequence (a_n) defined by:

$$a_n = 5a_{n-1} - 6a_{n-2} \quad \text{for } n \geq 2$$

with $a_0 = 1$, $a_1 = 3$.

Exercise 5. Find a recurrence relation for the number of ways to tile a $3 \times n$ board using 1×2 dominoes (which can be placed vertically as 2×1 or horizontally as 1×2).

Exercise 6. A robot needs to move from point A to point B on a $1 \times n (n > 4)$ grid. The robot can move with three types of jumps: 1 unit, 2 units, or 4 units forward. Let a_n be the number of different ways the robot can travel the distance of n units.

a) Establish a recurrence relation for a_n.

b) Calculate a_5.

Exercise 7. Find the general formula for the sequence (a_n) defined by:

$$a_n - 4a_{n-1} + 3a_{n-2} = 2^n \quad \text{for } n \geq 2$$

with initial conditions $a_0 = 1$, $a_1 = 2$.

Exercise 8. Given a non-negative integer N, count the number of different ways to arrange N opening brackets and N closing brackets correctly. A correct bracket arrangement can be defined recursively as follows:

- () is a correct bracket arrangement.
- If X is a correct arrangement, then ()X, (X), and X() are also correct bracket arrangements.

Hint: There is a bijection between the set of incorrect bracket strings (with N pairs) and the set of strings with $N + 1$ opening brackets '(' and $N - 1$ closing brackets ')'.

Exercise 9. For each of the following recurrence relations, provide an expression for the running time $T(n)$ if it can be solved using the Master Theorem. Otherwise, indicate that the Master Theorem does not apply.

a) $T(n) = 4T(\frac{n}{2}) + n^2$

b) $T(n) = T(\frac{n}{2}) + 2^n$

c) $T(n) = 2^n T(\frac{n}{2}) + n^n$

d) $T(n) = 2T(\frac{n}{2}) + \frac{n}{\log n}$

e) $T(n) = 0.5T(\frac{n}{2}) + \frac{1}{n}$
f) $T(n) = 16T(n/4) + n!$
g) $T(n) = 3T(\frac{n}{3}) + \sqrt{n}$
h) $T(n) = 3T(n/4) + n \log n$
i) $T(n) = 3T(\frac{n}{3}) + \frac{n}{2}$
j) $T(n) = 6T(\frac{n}{3}) + n^2 \log n$
k) $T(n) = 4T(\frac{n}{2}) + \frac{n}{\log n}$
l) $T(n) = 64T(n/8) - n^2 \log n$
m) $T(n) = 7T(\frac{n}{3}) + n^2$
n) $T(n) = 4T(\frac{n}{2}) + \log n$
o) $T(n) = T(\frac{n}{2}) + n(2 - \cos n)$